Using FaIR to Explore Future Global Warming Under SSP Scenarios

Comparison with IPCC AR6 projections and a simple emissions-perturbation experiment

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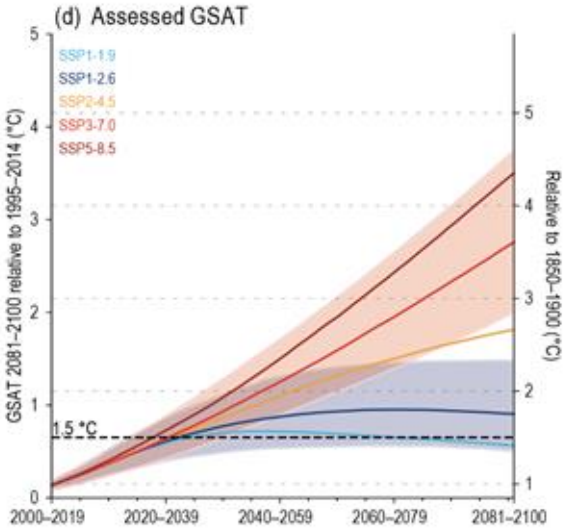
Goal

- use FaIR to simulate future global temperature under SSP scenarios
- Compare the FaIR output with IPCC AR6 Figure



- Test how warming changes when one emissions pathway is modified

FaIR workflow

- FaIR: “Finite Amplitude Impulse Response simple climate model”
- Dimensionality:

[time, scenario, config, specie, box/layer]

In years	emissions pathway	Climate parameter set	Forcings (CO2, CH4…)	Gas cycle & ocean layer

basic_run_example.ipynb

```
from fair import FAIR
f = FAIR()
```

→ create instance named ‘f’

```
from fair.io import read_properties
f.define_time(2000, 2050, 1)
f.define_scenarios(['abrupt', 'ramp'])
f.define_configs(['high', 'central', 'low'])
species, properties = read_properties()
f.define_species(species, properties)
f.ghg_method='Myhre1998'
```

→ define time, scenario, configuration, and species

```
f.allocate()
from fair.interface import fill
fill(f.emissions, 40, scenario='abrupt', specie='CO2 FFI')
...
f.run()
```

→ create empty FaIR, fill, run

https://docs.fairmodel.net/en/latest/intro.html

Experiments in `cmip6_ssp_emissions_run.ipynb`

Shared model setup: FaIR with **Thornhill (2021)** methane scheme; **1750–2100**, annual timestep; default full-climate species from read_properties(); scenario inputs from **RCMIP;** climate response configs from **CMIP6-calibrated** “4xCO2_cummins_ebm3.csv”

Experiment 1:

Baseline SSP runs

- ssp126, ssp245, ssp460, ssp585
- anomaly relative to **1986–2005**
- goal: reproduce class-style scenario divergence

Experiment 2:

Small perturbation

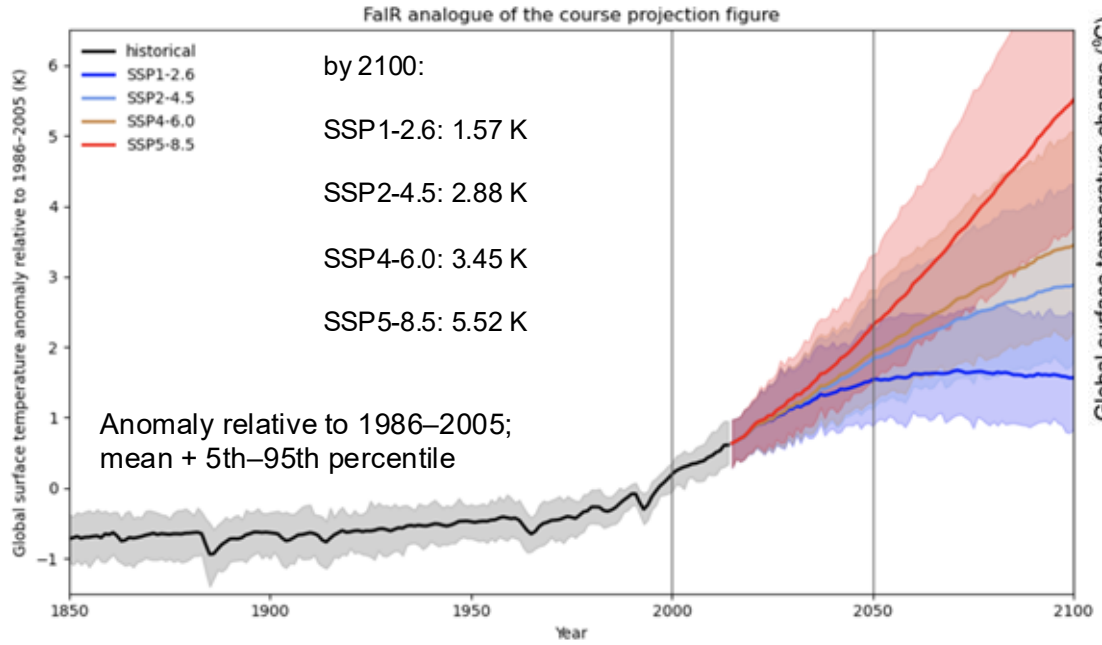
- start from ssp245
- reduce **CO2 FFI by 50% after 2030**
- goal: test sensitivity to one emissions change

Experiment 3:

AR6 comparison

- ssp119, ssp126, ssp245, ssp370, ssp585
- long-term warming = **2081–2100 relative to 1995–2014**
- goal: compare FaIR against **IPCC AR6 Figure 4.11(d)**

Experiment 1: Baseline run vs. Class figure



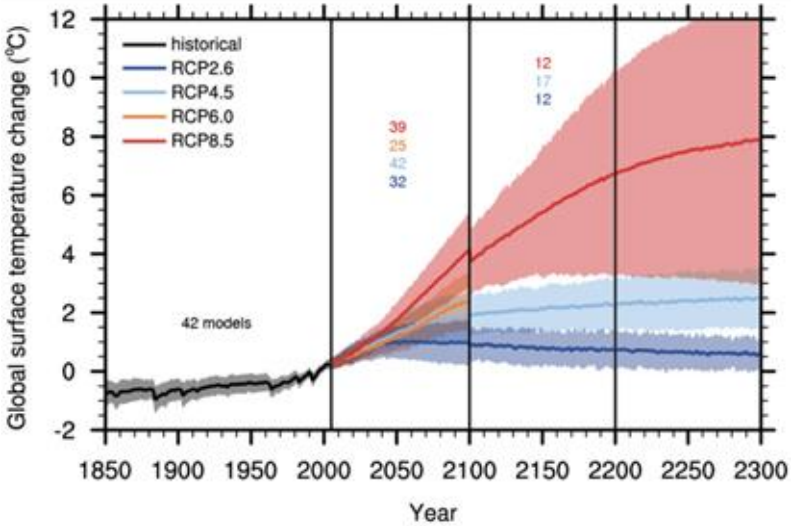
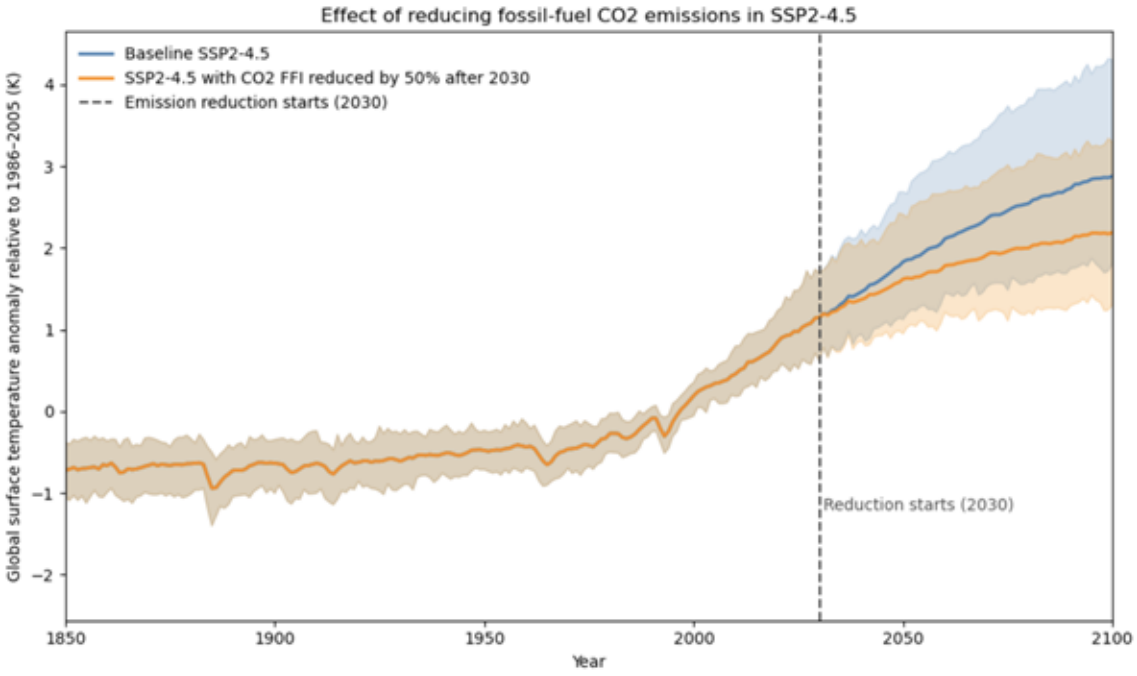


Figure 12.5: Time series of global annual mean surface air temperature anomalies (relative to 1986–2005) from CMIP5 concentration-driven experiments in IPCC AR5.

- shared historical pathway before divergence
- stronger forcing → stronger late-century warming
- uncertainty gets wider with time

- **SSP vs. RCP**
- ends at **2100** vs. extends farther
- percentile spread vs. sources of uncertainty

Experiment 2: Effect of reducing fossil-fuel CO2 emissions in SSP2-4.5



perturbation: CO_2 FFI × 0.5 after 2030

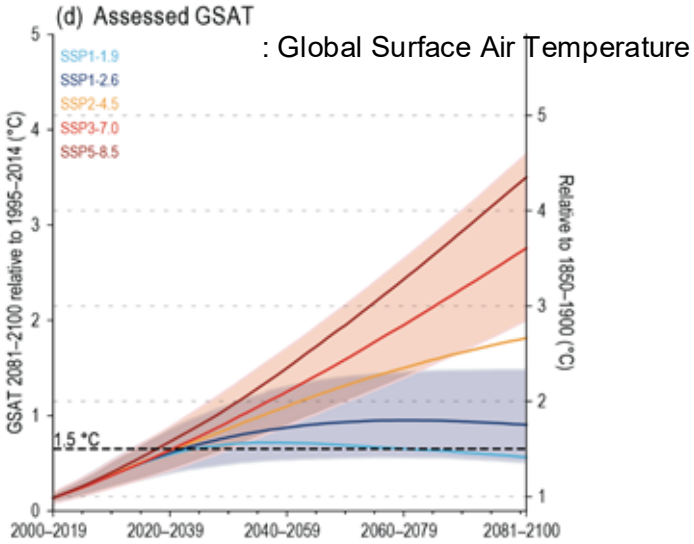
- isolate one emissions change

- test model sensitivity

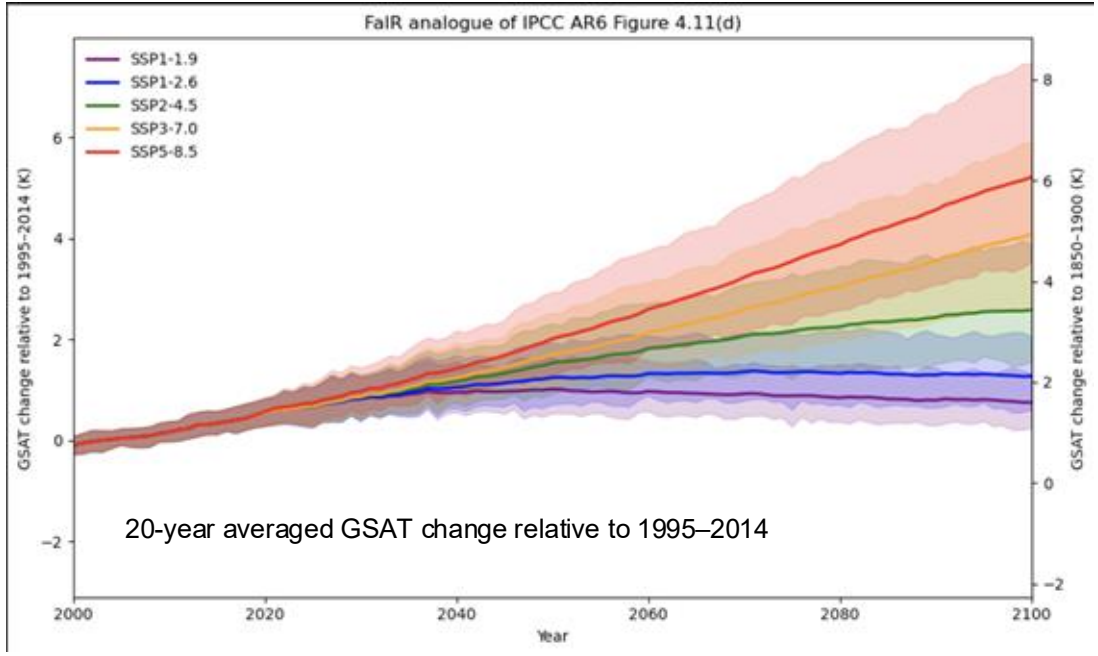
- same before 2030
- lower warming after 2030

- clear late-century cooling effect relative to baseline

Experiment 3: FaIR vs. AR6 (visual comparison)



AR6 Figure 4.11. The time series in **(d)** are constructed by taking the average of the constrained CMIP6 ranges and the emulator ranges. The y-axes on the right-hand side are shifted upward by 0.85°C, the central estimate of the observed warming for 1995–2014, relative to 1850–1900



- same scenario ordering
- similar late-century separation

- low-emissions → flatten more
- high-emissions → continue warming
- stronger forcing → wider spread uncertainty

Experiment 3: Quantitative comparison with IPCC AR6

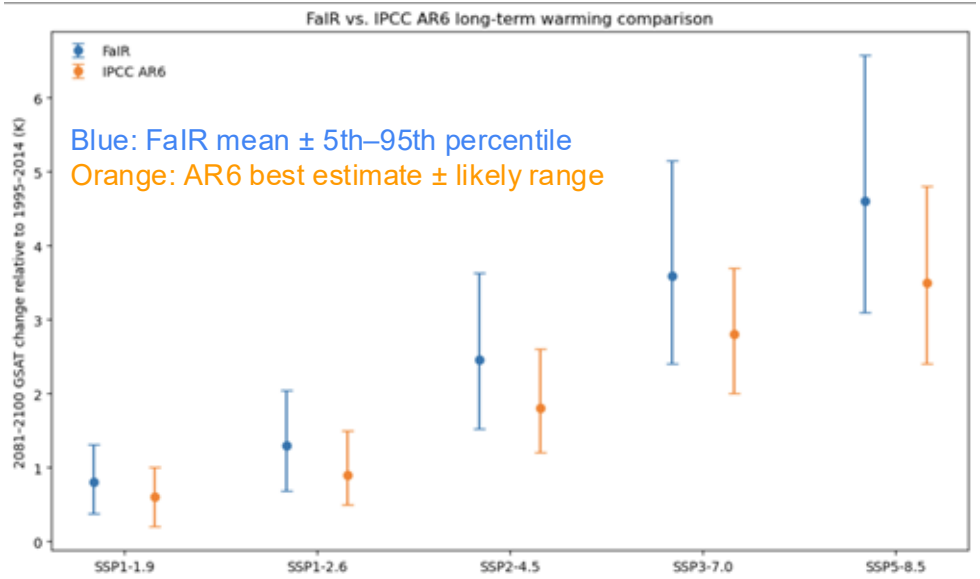


Table 4.5 from AR6. Assessed long-term GSAT change (2081–2100) based on multiple lines of evidence for comparison.

Comparison method:

- same 5 priority SSPs
- same 2081–2100 vs. 1995–2014 warming metric
- AR6 benchmark from Table 4.5

Result:

1)FaIR warmer than AR6 in all cases

2) Higher forcing → bigger difference

Why they are different?

- direct FaIR run vs. AR6 assessed values
- different uncertainty construction
- different overall workflow

Conclusions, limitations, and next steps

Key takeaways

- FaIR reproduced the main scenario-divergence pattern
- A simple CO_2 perturbation reduced late-century warming
- Agreement with AR6 was good overall
- My FaIR runs were warmer than AR6, especially at high forcing

Limits

- example notebook
- direct FaIR runs vs. AR6 assessed values
- FaIR spread vs. AR6 assessed range

Next steps

- match AR6 more closely
- investigate the warm bias
- test more CO_2 and non-CO_2 changes…